STIMULI FOR SPEECH ELICITATION

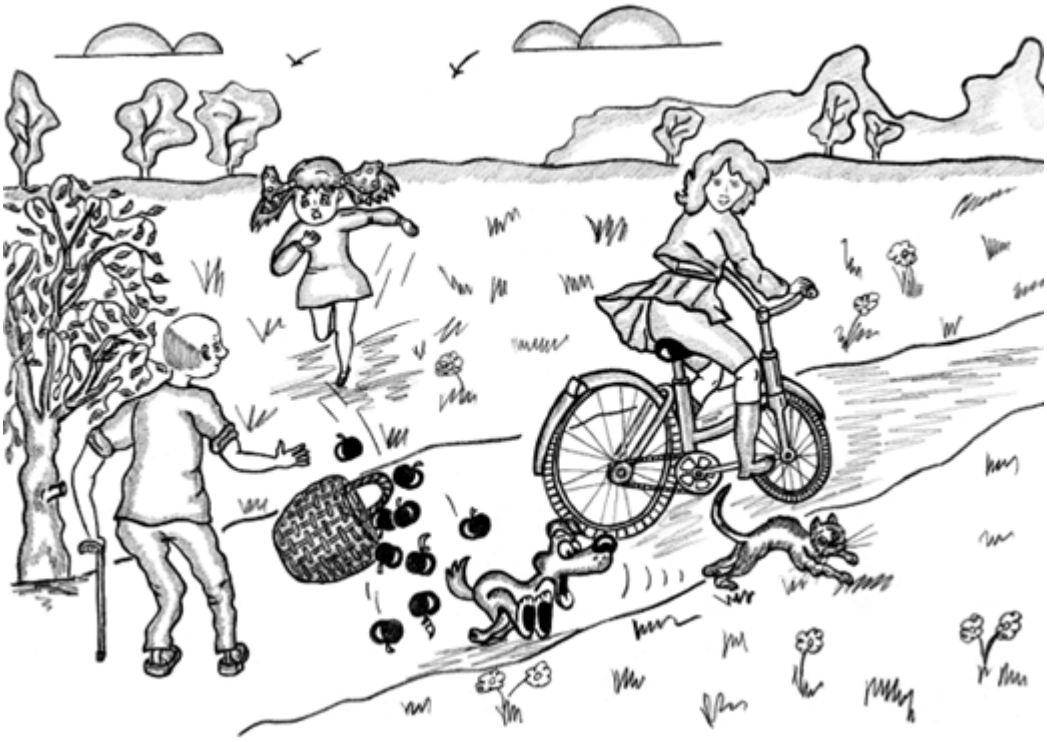


FIGURE 1. “THE BIKE” PICTURE

FIGURE 2. “THE ROBBERY” PICTURE

FEATURE EXTRACTION PARAMETERS

TABLE 1
FEATURE EXTRACTION PARAMETERS

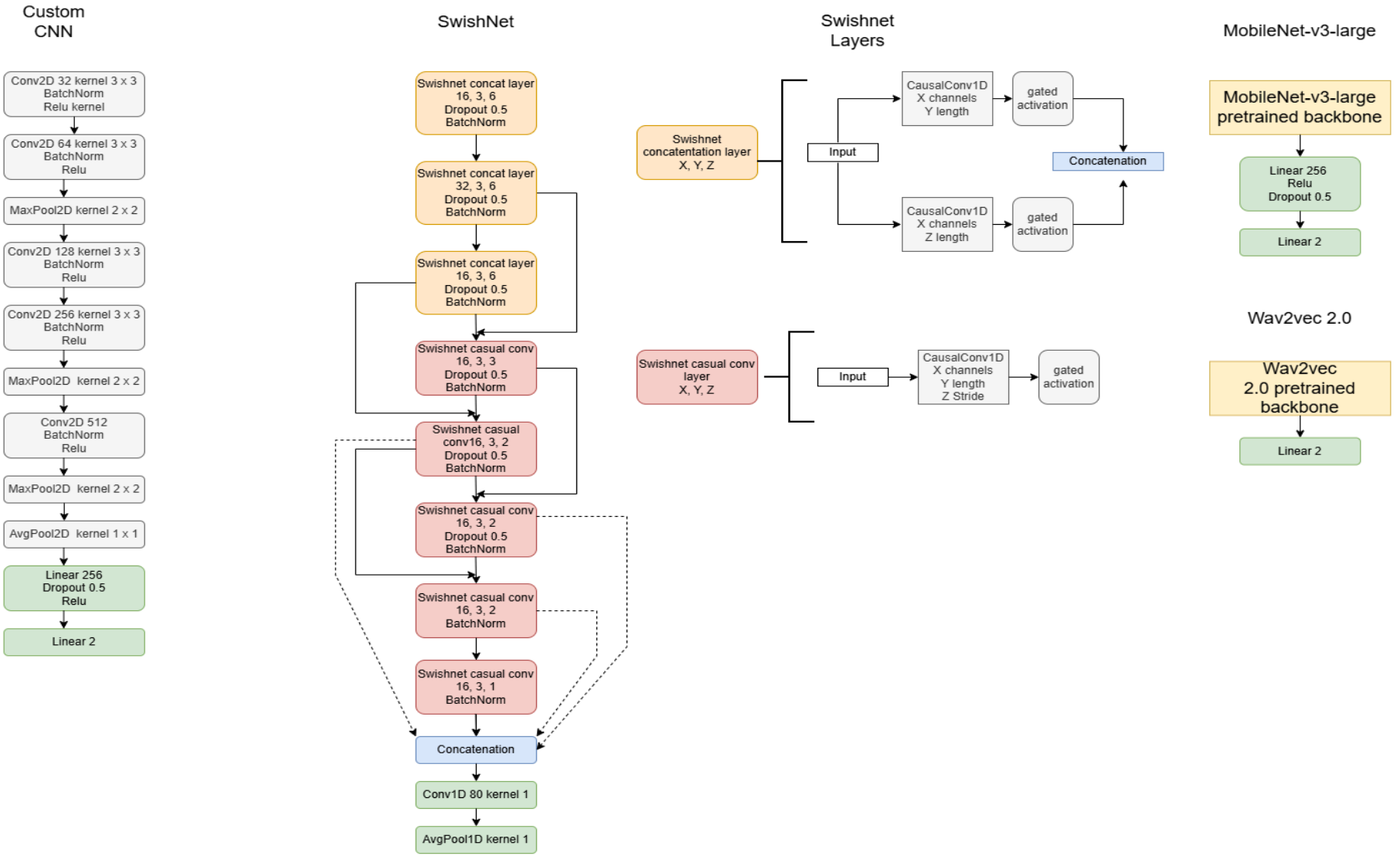
Feature	Hyperparameters	
MFCC	sample_rate	8000
	n_mfcc	64
	n_fft	20000
	win_length	10000
	hop_length	5000
	n_mels	200
Chroma	n_fft	20000
	win_length	10000
	hop_length	5000
	n_chroma	12
Spectral features: spectral_centroid spectral_bandwidth (p=2) spectral_contrast (fmin=50 Hz) spectral_flatness spectral_rolloff	n_fft win_length hop_length	20000 10000 5000
Zero Crossing Rate	frame_length	10000
	hop_length	5000

DEEP LEARNING MODELS TRAINING HYPERPARAMETERS

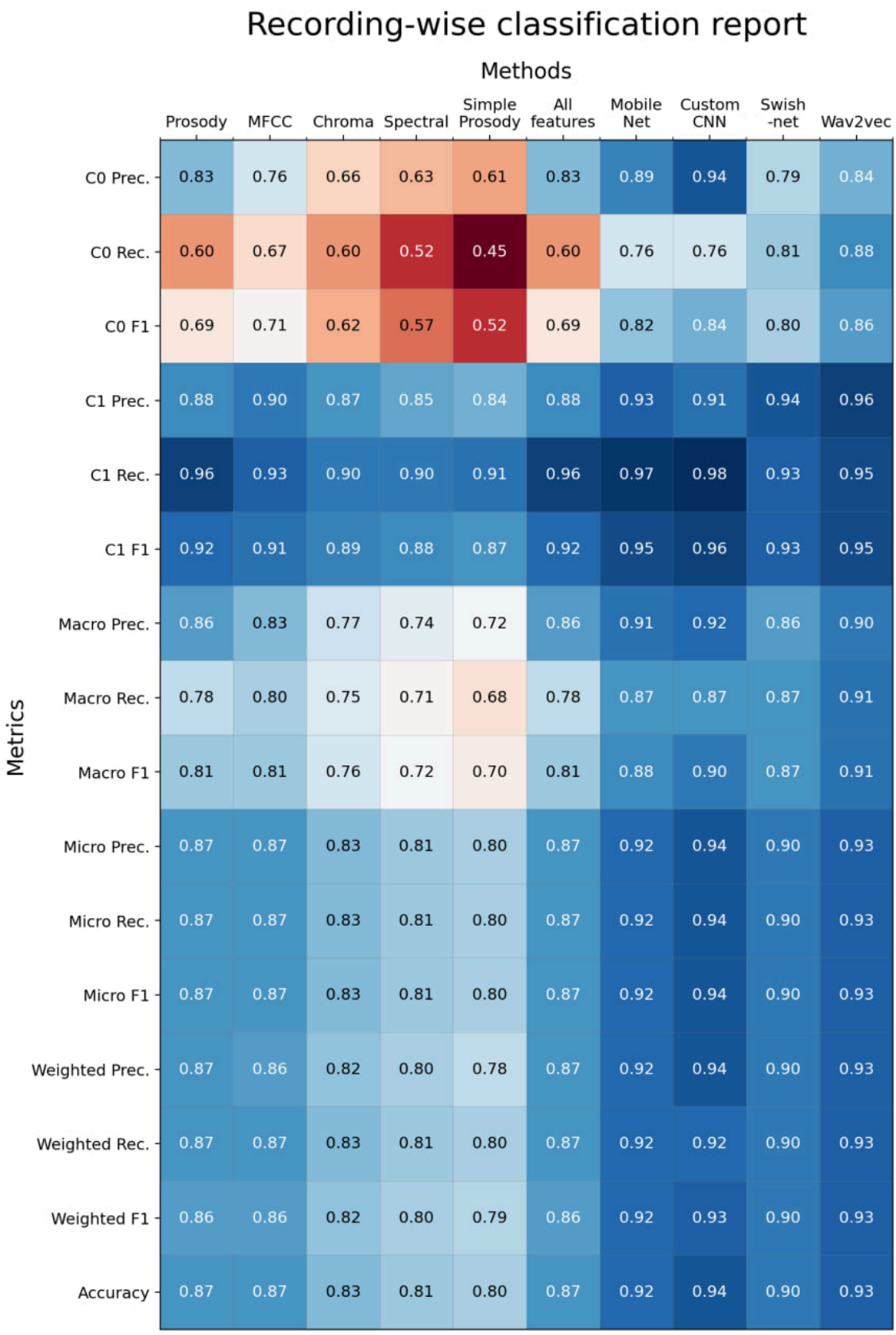
TABLE 2
DEEP LEARNING MODELS TRAINING HYPERPARAMETERS

Model	Training hyperparameters	
Custom CNN	Batch size	8
	Optimizer	AdamW (lr=1e-3, weight_decay=1e-4)
	Scheduler	CosineAnnealingLR (T_max=10, eta_min=1e-6)
Swishnet	MFCC	128 (n_mels=128, n_fft=512, win=512, hop=256)
	Batch size	64
	Weighted sampling according to class sizes	
	Optimizer	Adam (lr=$1e^{-3}$, weight_decay=$1e^{-5}$)
	Scheduler	ReduceLROnPlateau(factor=0.5, patience=3)
MobileNet-v3-large (fine tuning)	MFCC	128 (n_mels=128, n_fft=512, win=512, hop=256)
	Batch size	16
	Optimizer	Adam (lr=1e-3, weight_decay=1e-5)
	Scheduler	ReduceLROnPlateau
Wav2vec 2.0 (fine-tuning)	Batch size	16
	Optimizer	Adam (lr=1e-3, weight_decay=1e-5)

DEEP LEARNING ARCHITECTURES



DETAILED METRICS



Participant-wise classification report

Metrics \ Methods	Prosody	MFCC	Chroma	Spectral	Simple Prosody	All features	Mobile Net	Custom CNN	Swish -net	Wav2vec
C0 Prec.	0.88	0.87	0.87	0.69	0.87	1.00	0.89	0.94	0.95	0.86
C0 Rec.	0.71	0.62	0.62	0.43	0.62	0.67	0.81	0.76	0.86	0.90
C0 F1	0.79	0.72	0.72	0.53	0.72	0.80	0.85	0.84	0.90	0.88
C1 Prec.	0.89	0.86	0.86	0.80	0.86	0.88	0.92	0.91	0.94	0.96
C1 Rec.	0.96	0.96	0.96	0.92	0.96	1.00	0.96	0.98	0.98	0.94
C1 F1	0.92	0.91	0.91	0.85	0.91	0.94	0.94	0.94	0.96	0.95
Macro Prec.	0.89	0.86	0.86	0.74	0.86	0.94	0.91	0.93	0.95	0.91
Macro Rec.	0.84	0.79	0.79	0.68	0.79	0.83	0.89	0.87	0.92	0.92
Macro F1	0.86	0.81	0.81	0.69	0.81	0.87	0.90	0.89	0.93	0.92
Micro Prec.	0.89	0.86	0.86	0.78	0.86	0.90	0.92	0.92	0.94	0.93
Micro Rec.	0.89	0.86	0.86	0.78	0.86	0.90	0.92	0.92	0.94	0.93
Micro F1	0.89	0.86	0.86	0.78	0.86	0.90	0.92	0.92	0.94	0.93
Weighted Prec.	0.89	0.86	0.86	0.77	0.86	0.91	0.92	0.92	0.94	0.93
Weighted Rec.	0.89	0.86	0.86	0.78	0.86	0.90	0.92	0.92	0.94	0.93
Weighted F1	0.89	0.85	0.85	0.76	0.85	0.90	0.92	0.91	0.94	0.93
Accuracy	0.89	0.86	0.86	0.78	0.86	0.90	0.92	0.92	0.94	0.93